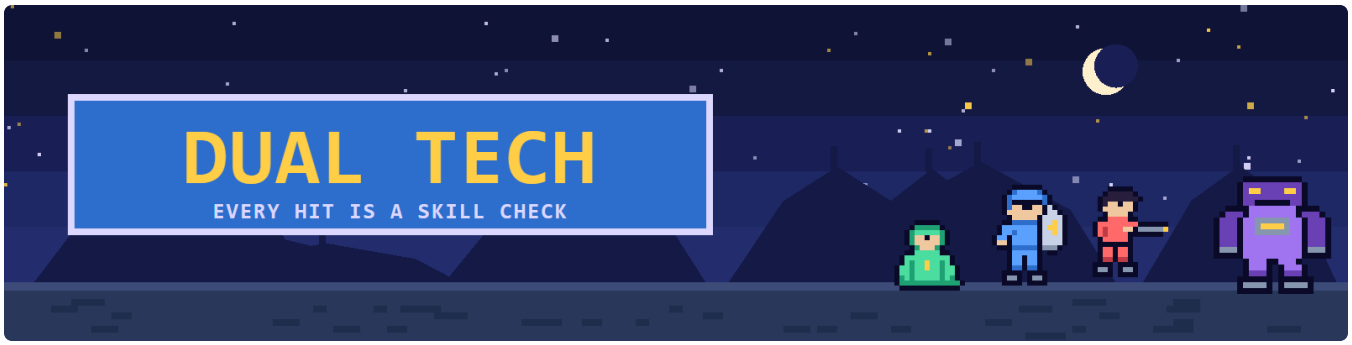
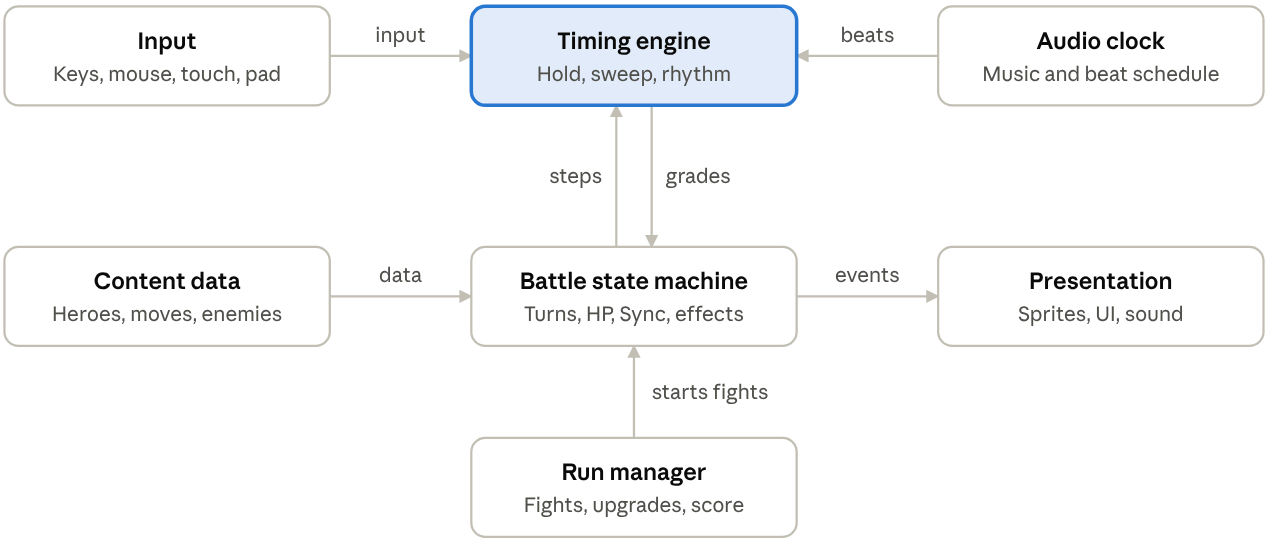
Dual Tech: technical design



Architecture

The game is small enough for one process and seven parts. The timing engine is the core: everything that touches input passes through it.

Every hit goes through one timing engine



architecture · 7 parts

The battle sends a list of timing steps and gets grades back. It never sees raw input. The presentation layer only listens for events (hit, heal, stagger, dual tech) and plays the animation and sound for each.

Battle state machine

A fight is one state machine, and only the Running state advances the clock. This gives wait-mode turn bars for free: the clock simply stops in every other state.

State	Clock	Enters when	Leaves to
Running	Turn bars fill	Fight starts, or an action finishes	Command (a hero's bar is full) or Enemy action (the enemy's bar is full)
Command	Paused	A hero is ready	Timing (a command is picked)
Timing	Paused	A command starts	Resolve (every hit is graded)
Resolve	Paused	Grades are in	Animating
Enemy action	Paused	The enemy's bar is full	Animating
Animating	Paused	An effect plays (about 0.6 s)	Running, or End if one side is down
End	Stopped	Enemy or party at 0 HP	Upgrade pick, or the results screen

Timing engine

Every timing hit, for any hero, runs through one engine with one interface. A command is a list of steps, and a dual tech is just a longer list. That's why chaining two styles costs no extra code.

- **Input:** the engine only receives `press(t)` and `release(t)`, with timestamps. Keyboard, mouse, touch and gamepad all map onto those two calls.
- **Step types:** `hold` (Brick), `sweep` (Vex), `rhythm` (Lumen). Each one implements `begin(t)`, `update(t)`, `press(t)`, `release(t)` and `draw(t)`, and returns a grade.
- **Output:** a list of results, one per step: `{grade: perfect | good | miss, quality: 0–1, hero}`. Rhythm steps also return a grade per beat.
- **Modifiers:** zone width (Amp, upgrades) and speed (upgrades, boss bells) are passed in with each step, never hard-coded in the step type.

Abilities never read input directly. They only see the results list, so you can test them without playing.

Timing precision

The perfect windows are tight (Lumen: 70 ms either side of the beat), so the timing code matters more than anything else.

- Grade inputs by the input event's timestamp, not by the frame it was processed on. A dropped frame should never turn a perfect into a good.
- Schedule Lumen's beats on the audio clock, so the beats and the music can't drift apart.
- Add a latency offset setting with a short calibration screen: tap along to 8 beats and store the average offset.
- Keep all windows in one tuning table (zone sizes, sweep speeds, charge rate, beat spacing). That way balance changes are data edits, not code changes.

Content as data

Heroes, abilities, dual techs, enemies, encounters and upgrades are data records. Each ability is a list of steps plus an effect function that turns results into damage, heals or status effects. Adding a dual tech means adding one record. Only Holy charge and Overclock volley need new code, because they run two inputs at the same time.

Testing

- **Simulated input:** a bot that presses at a chosen offset from each zone's centre. Use it to run a whole fight at 10x speed and check that it ends without errors.
- **Balance runs:** simulate players who always hit perfect, always good, or 50/50, and record how long each fight takes and how much HP is lost.
- **Unit tests** for grading at the window edges: exactly on the edge, one millisecond inside and one outside.

Build milestones

Each milestone ends in a gate. Don't start the next one until the gate passes.

1. **Core combat:** the prototype rebuilt on this architecture, with all 6 abilities, Sync and Amp. Gate: one fight is fun from start to finish.
2. **Dual techs and feel:** the 3 chained dual techs plus Breach, sounds for each hero, hit-stop and screen shake. Gate: playtesters can tell the three styles apart.

3. **Content:** 5 encounters, upgrade picks, run scoring, the boss's second phase. Gate: a full run is playable.
4. **Polish:** music on tempo, a balance pass, title and results screens, a release build. Gate: shipped.

Risks

- **Input latency across devices** can break the rhythm windows. Fix: calibration plus timestamp-based grading.
- **Switching styles mid-combo** may feel unfair. Fix: a lead-in of about 300 ms between steps, tuned in the data.
- **Simultaneous-input dual techs** need new step types. Fix: build them last, and cut them first if time runs short.

Cut list, in order

1. Holy charge and Overclock volley
2. The boss's second phase beyond the overheat
3. Upgrade picks (ship a fixed run)
4. Encounters 2 and 4 (the golem takes over the bell twist; see the GDD's encounter fallback)